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# STEP 1: Install Libraries
!pip install faiss-cpu transformers torch pandas

# STEP 2: Upload Dataset
from google.colab import files
uploaded = files.upload()

# STEP 3: Load Dataset
import pandas as pd

file_name = list(uploaded.keys())[0]
df = pd.read_csv(file_name)

print("Dataset Loaded Successfully")
print(df.head())

# STEP 4: Select Text Column
TEXT_COLUMN = df.columns[0]
item_texts = df[TEXT_COLUMN].astype(str).tolist()

print("Total items:", len(item_texts))
print("Sample item:", item_texts[0])

# STEP 5: Load Transformer Model
from transformers import AutoTokenizer, AutoModel
import torch
import numpy as np
import faiss

device = "cuda" if torch.cuda.is_available() else "cpu"

model_name = "sentence-transformers/all-MiniLM-L6-v2"
tokenizer = AutoTokenizer.from_pretrained(model_name)
model = AutoModel.from_pretrained(model_name).to(device)
model.eval()

# STEP 6: Encode Texts into Embeddings
def encode_texts(texts, batch_size=32):
    all_embeddings = []

    with torch.no_grad():
        for i in range(0, len(texts), batch_size):
            batch = texts[i:i+batch_size]

            tokens = tokenizer(
                batch,
                padding=True,
                truncation=True,
                return_tensors="pt"
            ).to(device)

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        outputs = model(**tokens)

        embeddings = outputs.last_hidden_state.mean(dim=1)
        embeddings = embeddings.cpu().numpy().astype("float32")

        all_embeddings.append(embeddings)

    return np.vstack(all_embeddings)

item_embeddings = encode_texts(item_texts)

# STEP 7: Normalize Embeddings
faiss.normalize_L2(item_embeddings)

print("Embeddings shape:", item_embeddings.shape)

# STEP 8: Build FAISS Index
embedding_dim = item_embeddings.shape[1]
index = faiss.IndexFlatIP(embedding_dim)

index.add(item_embeddings)

print("FAISS index built with", index.ntotal, "items")

# STEP 9: Similarity Search Function
def get_similar_items(item_id, k=5):
    query_vector = item_embeddings[item_id:item_id+1]

    scores, indices = index.search(query_vector, k)

    return scores[0], indices[0]

# STEP 10: Test Similarity Retrieval
query_id = 0

scores, idxs = get_similar_items(query_id, k=5)

print("\nQUERY ITEM:")
print(item_texts[query_id])

print("\nSIMILAR ITEMS:")
for score, idx in zip(scores, idxs):
    print(f"-> {item_texts[idx]} (score={score:.3f})")

```

Collecting faiss-cpu

Downloading faiss_cpu-1.13.2-cp310-abi3-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadata (7.6 kB)
Requirement already satisfied: transformers in
/usr/local/lib/python3.12/dist-packages (5.0.0)
Requirement already satisfied: torch in

/usr/local/lib/python3.12/dist-packages (2.10.0+cpu)
Requirement already satisfied: pandas in
/usr/local/lib/python3.12/dist-packages (2.2.2)
Requirement already satisfied: numpy<3.0,>=1.25.0 in
/usr/local/lib/python3.12/dist-packages (from faiss-cpu) (2.0.2)
Requirement already satisfied: packaging in
/usr/local/lib/python3.12/dist-packages (from faiss-cpu) (26.0)
Requirement already satisfied: filelock in
/usr/local/lib/python3.12/dist-packages (from transformers) (3.24.3)
Requirement already satisfied: huggingface-hub<2.0,>=1.3.0 in
/usr/local/lib/python3.12/dist-packages (from transformers) (1.5.0)
Requirement already satisfied: pyyaml>=5.1 in
/usr/local/lib/python3.12/dist-packages (from transformers) (6.0.3)
Requirement already satisfied: regex!=2019.12.17 in
/usr/local/lib/python3.12/dist-packages (from transformers)
(2025.11.3)
Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in
/usr/local/lib/python3.12/dist-packages (from transformers) (0.22.2)
Requirement already satisfied: typer-slim in
/usr/local/lib/python3.12/dist-packages (from transformers) (0.24.0)
Requirement already satisfied: safetensors>=0.4.3 in
/usr/local/lib/python3.12/dist-packages (from transformers) (0.7.0)
Requirement already satisfied: tqdm>=4.27 in
/usr/local/lib/python3.12/dist-packages (from transformers) (4.67.3)
Requirement already satisfied: typing-extensions>=4.10.0 in
/usr/local/lib/python3.12/dist-packages (from torch) (4.15.0)
Requirement already satisfied: setuptools in
/usr/local/lib/python3.12/dist-packages (from torch) (75.2.0)
Requirement already satisfied: sympy>=1.13.3 in
/usr/local/lib/python3.12/dist-packages (from torch) (1.14.0)
Requirement already satisfied: networkx>=2.5.1 in
/usr/local/lib/python3.12/dist-packages (from torch) (3.6.1)
Requirement already satisfied: jinja2 in
/usr/local/lib/python3.12/dist-packages (from torch) (3.1.6)
Requirement already satisfied: fsspec>=0.8.5 in
/usr/local/lib/python3.12/dist-packages (from torch) (2025.3.0)
Requirement already satisfied: python-dateutil>=2.8.2 in
/usr/local/lib/python3.12/dist-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in
/usr/local/lib/python3.12/dist-packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in
/usr/local/lib/python3.12/dist-packages (from pandas) (2025.3)
Requirement already satisfied: hf-xet<2.0.0,>=1.2.0 in
/usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0->transformers) (1.3.1)
Requirement already satisfied: httpx<1,>=0.23.0 in
/usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0->transformers) (0.28.1)
Requirement already satisfied: typer in

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/usr/local/lib/python3.12/dist-packages (from huggingface-  
hub<2.0,>=1.3.0->transformers) (0.24.1)  
Requirement already satisfied: six>=1.5 in  
/usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2-  
>pandas) (1.17.0)  
Requirement already satisfied: mpmath<1.4,>=1.1.0 in  
/usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch)  
(1.3.0)  
Requirement already satisfied: MarkupSafe>=2.0 in  
/usr/local/lib/python3.12/dist-packages (from jinja2->torch) (3.0.3)  
Requirement already satisfied: anyio in  
/usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0-  
>huggingface-hub<2.0,>=1.3.0->transformers) (4.12.1)  
Requirement already satisfied: certifi in  
/usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0-  
>huggingface-hub<2.0,>=1.3.0->transformers) (2026.2.25)  
Requirement already satisfied: httpcore==1.* in  
/usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0-  
>huggingface-hub<2.0,>=1.3.0->transformers) (1.0.9)  
Requirement already satisfied: idna in /usr/local/lib/python3.12/dist-  
packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.3.0-  
>transformers) (3.11)  
Requirement already satisfied: h11>=0.16 in  
/usr/local/lib/python3.12/dist-packages (from httpcore==1.*-  
>httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.3.0->transformers) (0.16.0)  
Requirement already satisfied: click>=8.2.1 in  
/usr/local/lib/python3.12/dist-packages (from typer->huggingface-  
hub<2.0,>=1.3.0->transformers) (8.3.1)  
Requirement already satisfied: shellingham>=1.3.0 in  
/usr/local/lib/python3.12/dist-packages (from typer->huggingface-  
hub<2.0,>=1.3.0->transformers) (1.5.4)  
Requirement already satisfied: rich>=12.3.0 in  
/usr/local/lib/python3.12/dist-packages (from typer->huggingface-  
hub<2.0,>=1.3.0->transformers) (13.9.4)  
Requirement already satisfied: annotated-doc>=0.0.2 in  
/usr/local/lib/python3.12/dist-packages (from typer->huggingface-  
hub<2.0,>=1.3.0->transformers) (0.0.4)  
Requirement already satisfied: markdown-it-py>=2.2.0 in  
/usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer-  
>huggingface-hub<2.0,>=1.3.0->transformers) (4.0.0)  
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in  
/usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer-  
>huggingface-hub<2.0,>=1.3.0->transformers) (2.19.2)  
Requirement already satisfied: mdurl~=0.1 in  
/usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0-  
>rich>=12.3.0->typer->huggingface-hub<2.0,>=1.3.0->transformers)  
(0.1.2)  
Downloading faiss_cpu-1.13.2-cp310-abi3-  
manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl (23.8 MB)
```

0:00:00

<IPython.core.display.HTML object>

Saving amazon_categories.csv to amazon_categories.csv

Dataset Loaded Successfully

	id	category_name
0	1	Beading & Jewelry Making
1	2	Fabric Decorating
2	3	Knitting & Crochet Supplies
3	4	Printmaking Supplies
4	5	Scrapbooking & Stamping Supplies

Total items: 248
Sample item: 1

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
```

The secret `HF\_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (<https://huggingface.co/settings/tokens>), set it as secret in your Google Colab and restart your session.
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.

```
warnings.warn(
Warning: You are sending unauthenticated requests to the HF Hub.
Please set a HF_TOKEN to enable higher rate limits and faster
downloads.
```

```
WARNING:huggingface_hub.utils._http:Warning: You are sending
unauthenticated requests to the HF Hub. Please set a HF_TOKEN to
enable higher rate limits and faster downloads.
```

```
{"model_id": "16b0c1044ece404ab20bcf4e732d0569", "version_major": 2, "version_minor": 0}
```

```
{"model_id": "1858252301a64f378f40936af37107b1", "version_major": 2, "version_minor": 0}
```

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{"model_id": "2568b4167cee451bb3f9f3a31f3dbdc6", "version_major": 2, "version_minor": 0}
```

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{"model_id": "4ddd7c754eb942bca9dedb7e699dc083", "version_major": 2, "version_minor": 0}
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{"model_id": "e392f1d3832e4d70918719ce9f5a45c2", "version_major": 2, "version_minor": 0}
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{"model_id": "bcae6947bf21460990840ebbdd31204", "version_major": 2, "version_minor": 0}
```

```
{"model_id": "b0c0a89dbb5e41788a9e5d0229cd018d", "version_major": 2, "version_minor": 0}
```

BertModel LOAD REPORT from: sentence-transformers/all-MiniLM-L6-v2

Key	Status		
-----+-----+---+			
embeddings.position_ids	UNEXPECTED		

Notes:

- UNEXPECTED : can be ignored when loading from different task/architecture; not ok if you expect identical arch.

Embeddings shape: (248, 384)

FAISS index built with 248 items

QUERY ITEM:

1

SIMILAR ITEMS:

- > 1 (score=1.000)
- > 2 (score=0.805)
- > 3 (score=0.791)
- > 4 (score=0.769)
- > 7 (score=0.658)